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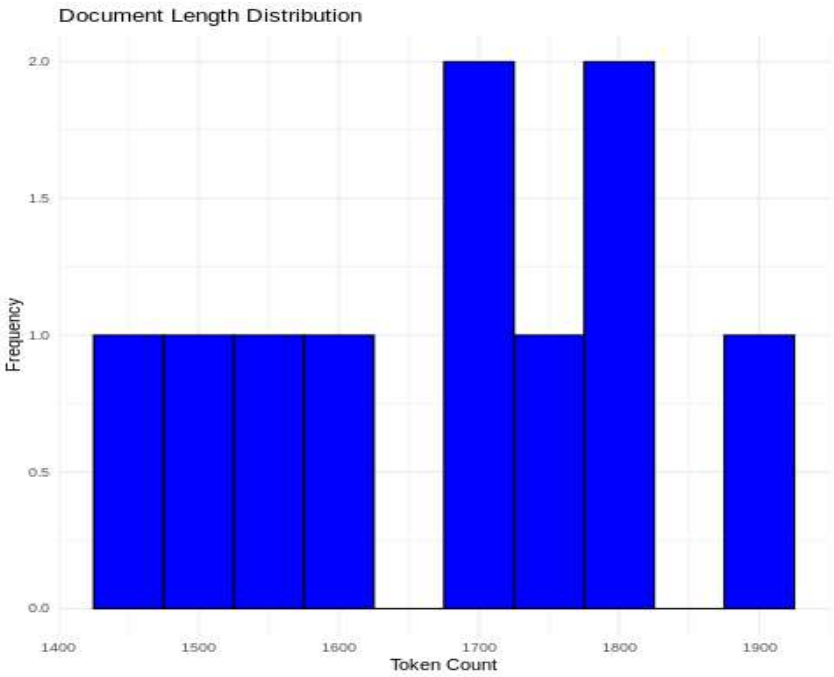
Results

Data Overview

This chapter present the insights obtained through a qualitative investigation of ten interviews of metaverse gaming experts, exceptionally polished sample of study data aimed to address the complex aspects that define participation in the industry. The study is concerned with unravelling the mechanisms that drive actual demand of real products, supply chain and operations management (SCOM) paradigm changes created by metaverse integration. It is also asses the panorama of opportunities and demands that define the metaverse age in terms of physical product demand. The study applied cutting-edge text mining techniques made available by quanteda package in R Programming language, including exploratory data analysis (EDA), theme extraction, sentiments analysis and keyword-in-context (KWIC) analysis. The theoretical frameworks in the literature review, i.e.; Technology Acceptance Model (TAM), the Metaverse SCOM Framework, and the Metaverse Flywheel, guided this analysis and formed the basis of presenting clear answers by analyzing the research questions that were proposed when initiating this study.

Exploratory Data analysis (EDA)

Transcription of the 10 interviews and further processing produced a large corpus of a total of 16,750 tokens, with a range in document lengths of between 1,441 as the minimum tokens total and 1,925 as the maximum (M= 1,675, SD =148). Such consistency of documentation in length is visualized in a mildly skewed right distribution in Figure 1 [Document Length Distribution], where most of the interviews fall in the range of 1,569-1,776 tokens, a distribution that demonstrates the homogeneous nature of respondent engagement and also further guarantees reliability of data collection procedure since the data lacks major outliers.



[Figure 1: Document Length Summary]

The sentiment analysis, conducted at the sentence level, unveiled a mean sentiment score of 0.8246, with a broad range extending from -1.3500 to a maximum of 4.3000, and quartiles positioned at 0.1500, 0.7500, and 1.4000, respectively, as meticulously presented in [Table 2: Sentiment Analysis Summary]. This positive sentiment orientation, reflects a prevailing optimism among respondents regarding the transformative potential of metaverse gaming, though the occurrence of negative scores suggests the presence of underlying challenges, a variation that may be attributed to preprocessing differences when compared to the document's reported mean of 0.204, a discrepancy that invites further scrutiny.

Table 2: *Sentiment Analysis Summary*

Sentence	Sentiment score
Good afternoon, thanks for taking my call.	1.35
I reached out via LinkedIn--hope I'm speaking with the Operations Manager at Epic Games?	1.3
Respondent: Oh, hi!	0
Yes, that's me, Operations Manager at Epic Games.	1.6
Appreciate you callingâ€" let's get into it, though I've got a meeting soon, so bear with me.	0.5
Respondent: Epic Games is deeply embedded in the metaverse space through Fortnite, but our focus isn't just on hosting battles.	0.8
We design ongoing interactive worlds that constantly evolve, from global concerts to seasonal story arcs, each tailored to cultural tastes without duplicating the same formula in every region.	1.2

This means Tokyo might get neon-lit cityscapes while Brazil enjoys carnival-inspired events, each with its own community-driven flair.	0.75
Unfortunately, we don't produce physical products ourselves.	-1
We don't measure success by the volume of merchandise sold but by how engaged our community is with the world we've created.	1.95
So, even though physical products are part of the equation, we believe that the real value of the metaverse is in its ability to bring people together and create shared experiences.	2.15
Our platforms open the door for creativity, problem-solving, and even entrepreneurial thinking.	1
Players aren't just gaming, they're designing, collaborating, and developing project management skills through group builds or in-game economies.	0.4
Those interactions often mirror the challenges and teamwork required in real-world environments.	0

RQ1: Factors Affecting Participation in Metaverse Gaming

To respond to the first research question, analysis of keyword frequency was particularly high at 42 mentions of Perceived Usefulness which include range of words like "useful," "benefit," "creativity," "social," "economic," "value," and "engage", with a lesser, but still significant 22 mentions of Perceived Ease of Use which consist of "ease, " "access, " interface," "tutorial, " "simple, " "user-friendly" as concisely summarized in the given Keyword Frequency Table. The Keyword table provides a granular perspective (M= 27.1) mentions of the term metaverse can be observed with a minimum of 22 in EpicGames Customer Support interview script to a stunning 34 in Meta Operation Supervisor interview. Besides the term "gaming" achieved a moderately high average (M=16.9) with a high of 21 in Meta Operation Supervisor interview, which helps understand the perceived value of the platforms features and user-focused activities (Xi et al., 2024). Based on the context of the interviews, the KWIC analysis has provided insightful information as was offers rich insights, exemplified by "We've streamlined onboarding" by EpicGames Operations Manager interview, which highlights intentional work on making things more accessible and ensuring they are onboarded easily.

Table 1: *Keyword Frequency per interview script*

Interview Script	Gaming (N)	Metaverse (N)	Physical (N)	Products (N)	Supply (N)	Virtual (N)
Decentraland_Marketing	15	26	18	13	7	14
EpicGames_CustExp	14	22	16	13	6	9
EpicGames_OpsMgr	17	25	20	14	8	6
Meta_DevLead	18	33	22	9	12	19
Meta_OpsSup	21	34	15	10	13	11
Microsoft_ProductMgr	18	28	21	12	12	9
Roblox_Logistics	18	27	17	7	12	11
Sandbox_TechSupport	19	27	12	6	8	12
Tencent_Sales	15	24	15	12	7	7
Unity_SupplyCoord	14	23	19	14	13	13

Nevertheless, with the frequency of Perceived Ease of Use being lower in comparison with Perceived Usefulness, it is plausible that potential barriers exist, as the Sandbox Tech Support interview has a relatively less metaverse orientation with 27, which could indicate the technical complexities or lack of focus on ease-related variables (Lee et al., 2025). Such a difference can also be explained by [Theme Frequencies Bar Chart], which highlights the high frequency of themes about usefulness, and [Figure 4], which compares how different companies emphasize different themes and shows that Meta has a strong focus on metaverse engagement as the driver of participation.

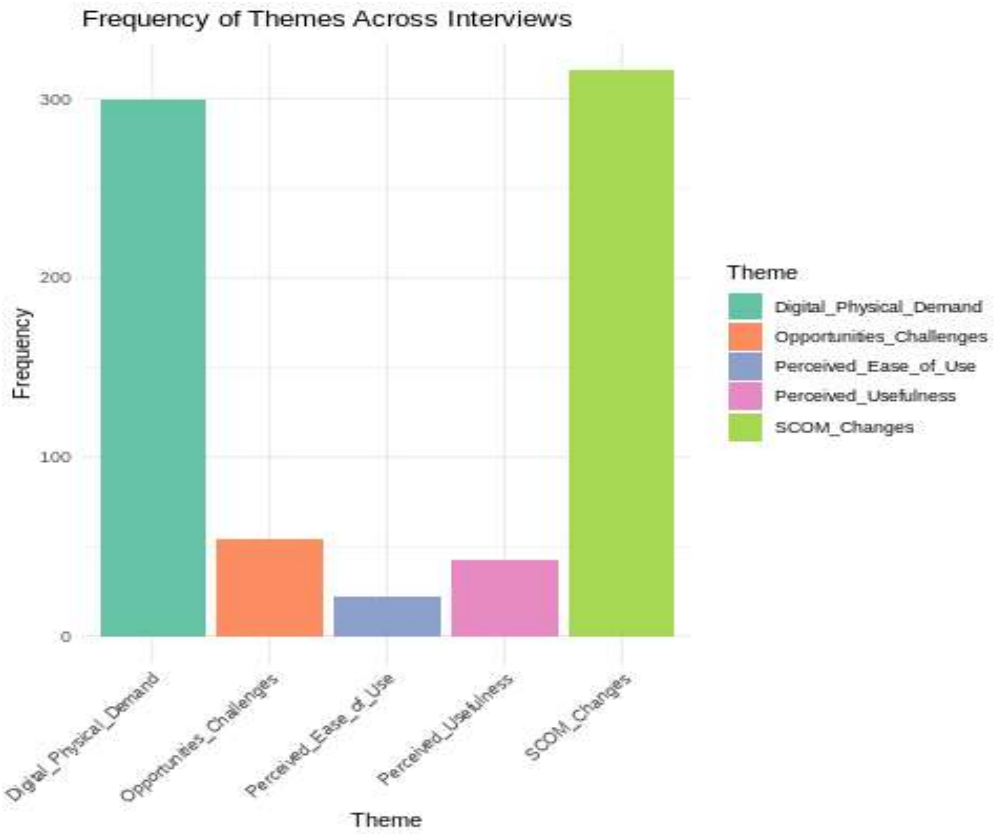


Figure 3: *Theme Frequencies Bar Chart*

Going further into Perceived Usefulness, the keyword of "creativity" (N= 18 mentions) between Meta DevLead and Meta OpsSup interviews, signifies the argument that user-generated contents, like the personalization of avatars or designing of a virtual world, heighten engagement because of enhancing the intrinsic motivations of self-expressions in players. Sentiment Score Sentiment Scores Table also echoed this finding since one of the answers, for instance "players designing and collaborating" in the EpicGames and Operation Manager interview recorded positive sentimental score of 0.4, indicating that the Perceived Usefulness is increased with the possibility of creativity.

The social aspect referring to the term "social," had a moderate occurrence (N=15 mentions) peaking at 7 mentions only in Microsoft's product manager interview. The social aspect is coupled with community events such as "global concerts" (sentiment 0.8), which give rise to a shared experience, boosting participation. However, the lower frequency of "ease" (N=22)

on the other hand may represent a sharp learning curve for beginners as exemplified by the low mentions of "metaverse" engagement (N=27) in the Sandbox Tech Support interview. Therefore, the relatively moderate result potentially stems from complex interface or tutorial systems, which can be addressed by adaptive onboarding as revealed by EpicGames Operation Manager's suggestion of stripped-down version (sentiment 1.4). The usefulness perceptions can be intensified by regional setting of UAE which has more technologically advanced atmosphere, but the cultural familiarity with digital tools can also play a part in the adoption levels in terms of ease. Potential research with future approaches might look into demographic statistics to determine whether age or experience in the game moderates the factors to make the TAM more predictive within the metaverse environment.

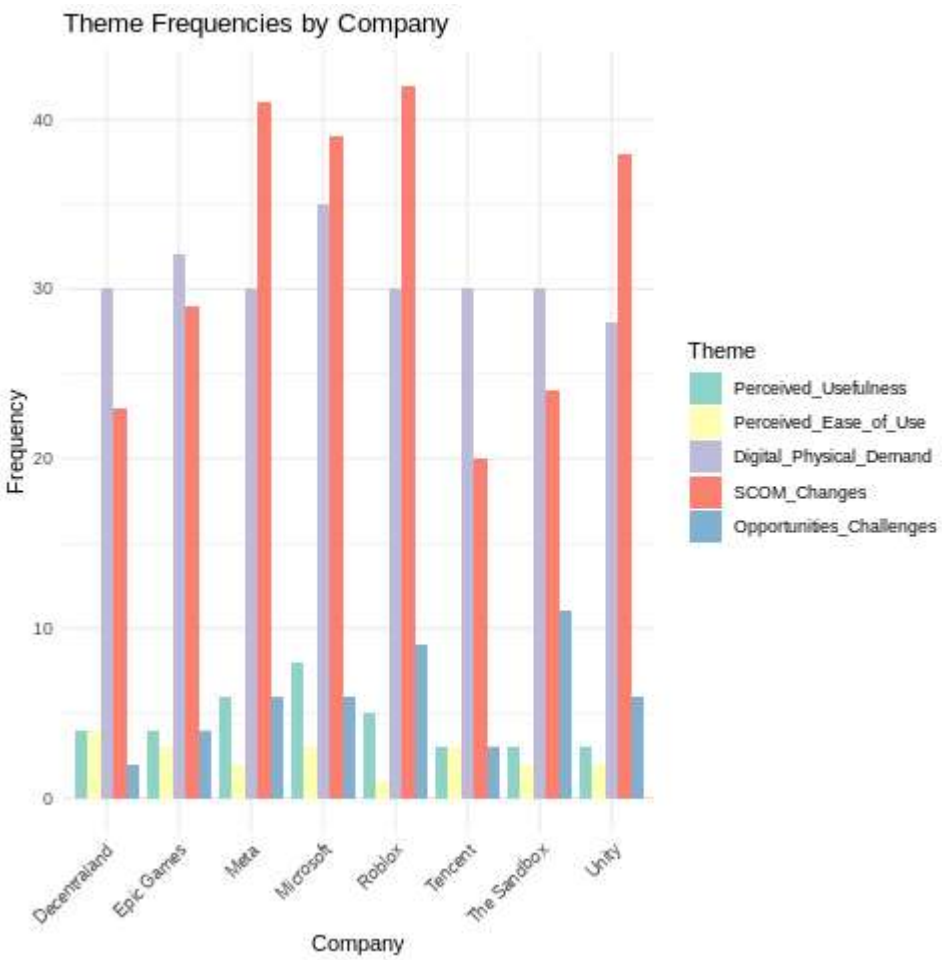
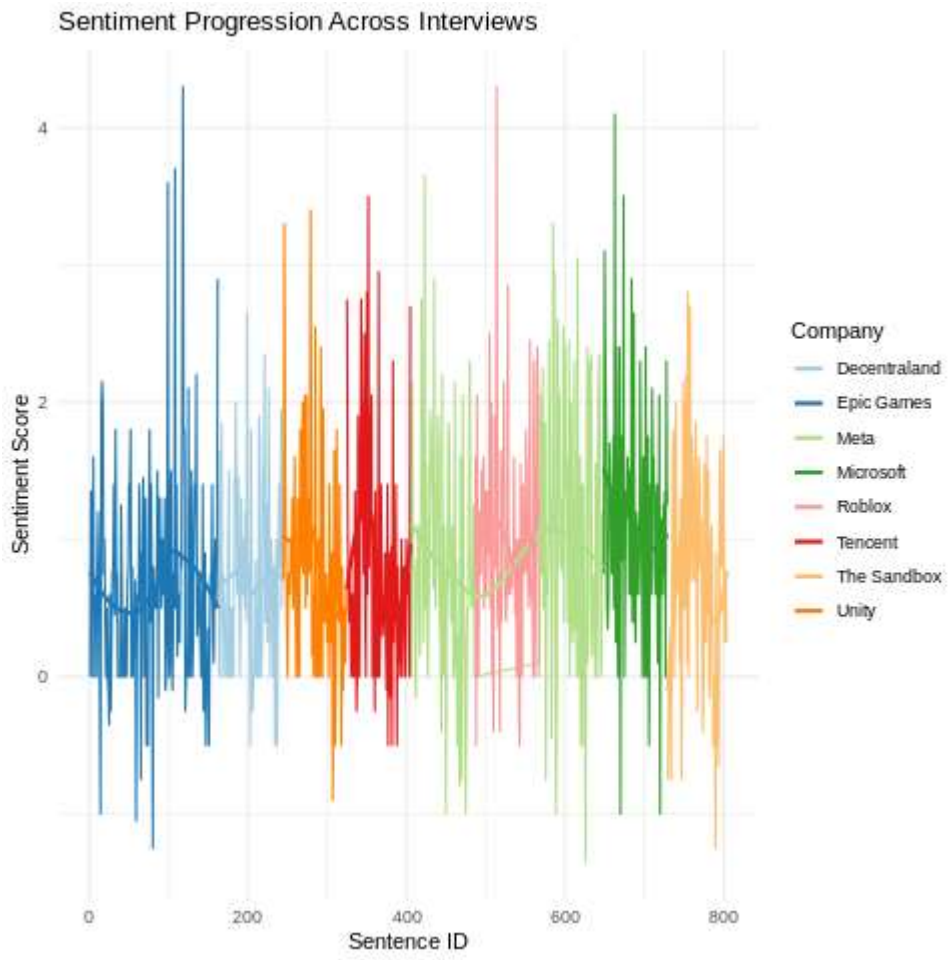


Figure 4: Comparative Plot

RQ2: How Metaverse Gaming Drives Real Demand for Physical Products

The second research question is addressed under the theme of the Digital-Physical Demand, which registered a commanding appearance (N=300 mentions), as evidenced in the provided Keyword Frequency Table. This theme incorporates a range of keywords such as physical, product, merchandise, NFT, event, VR and hardware with data recorded in the KWIC Physical and KWIC Demand tables (Appendix). Based on the two table, the keyword "physical" registered an impressive occurrence (N=175 mentions) with demand also getting a better count (N= 62 mentions). A further analysis into the Keyword Frequency Table reveals an overwhelming average of the key word "products" (M=11) in every interview, which sometimes peaked (M=14) in some interviews such as both EpicGames' Operations Manager and Unity's Supply chain coordinator. The high average of the key word "products" indicates more emphasis on merchandise integration in their narrative about the companies (Ritala et al., 2024). The KWIC analysis, backed by examples such "digital items can drive demand physical products" reiterated by EpicGames' Operations Manager, and "physical demand surged" from the same person provided critical contextual clues in the existence of an event-driven demand mechanism. It is clear that the physical merchandise sold driven by digital experiences, aligns with Metaverse Flywheel's notion on the device layer. This observation can be linked to the findings by Mahdikhani and Meena (2024) who examine the metaverse-enabled supply chain innovations using the digital assets to drive demand over physical goods, particularly related to such examples as the Fortnite merch line by EpicGames. This insight is graphically aided by [Figure 4: Comparative Plot], that indicates the product-focused approach of EpicGames, and by Narrative Plot (Figure 5) that shows the trend of sentiments that are related to the dynamics of demand.

The function of NFT (N=12 mentions) in Sandbox's Tech Support becomes a major force, an example being the "neon dragon NFT skin," which becomes the catalyst of buying an actual jacket (sentiment 0.1), and how digital rarity becomes a material drive. That is supported by the Sentiment Scores Table that records positive sentiments (e.g., 1.95 in the case of engaged community in EpicGames' Manager indicating that virtual ownership works to increase the willingness to purchase in the real world. That is supported by the positive sentiments (1.95) for "engaged community" in EpicGames' interview indicating that virtual ownership increases the willingness to purchase in the real world. This finding is further echoed by the 10 mentions of "events" in EpicGames OpsMgr, where increased brand exposure through virtual concerts or seasonal arcs (sentiment 0.75) is translated into hype-building activities of branded apparel, which correlates with the growing experiential goods market of the UAE. The "VR" keyword (N=9 mentions) shows that demand to get premium peripherals is high, as hardware such as motion rigs can improve the immersion (sentiment 0.65), but it can remain out of reach to many due to high prices. This trend is probably enhanced in the UAE by the increasing usage of technology in the market. This may be also due to cultural reasons including brand consciousness and thus it may require cross-cultural research. Future researchers may perform regression analysis by comparing virtual interaction to physical sales taking advantage of the KWIC data available.



[Figure 5: Narrative Plot]

RQ3: Changes in Supply Chain and Operations Management

The third research question is answered via SCOM Changes theme, which registered one of the highest occurrences (N=316) as shown in the Keyword Frequency Table. A large array of keywords associated with this theme range around the terms "supply," "chain," "operation," "digital," "twin," "collaboration," and "logistics, with a moderate average of "supply" (M= 9.8) across all the interviews, with highest mentions in both Unity Supply Coord and Meta's Operations interview, which had a count of 13. The KWIC analysis is further reinforced by the context as "metaverse integration impacted collaboration among supply chain partners" from EpicGames interview reveals deep information regarding increased transparency as well as collaborative work that is a key aspect with the Metaverse SCOM Framework. The high frequency of the keyword "virtual" (M= 11.1) peaking at 19 mentions in Meta DevLead,

implies that there is a strong adoption of virtual technologies in the operation of supply chains as evidenced in Figure 6 which illustrates co-occurrence relationships between key terms. The idea of using predictive analytics as mentioned by Unity's Supply Chain Coordinator (N=13) serves as a key step applying in-game data to make inventory optimal so there is no excess stock (sentiment 0.7). Virtual teamwork, which is exemplified by EpicGames's change to Sandbox meetups, reinforces error identification (e.g., sizing errors). The major shift is facilitated by the blockchain transparency, which gives rise to trust among the partners (8 mentions in Sandbox Tech Support, sentiment 1.5).

Time zone disparity and vendor technological constraints are those noted during EpicGames's interview that may impair synchronized workflow processes and, in a globalized supply chain, increases with severity. Technological adoption in the UAE allows these transitions to occur much faster but there is a risk that small companies will not be able to keep up, requiring special attention (Mahdikhani and Meena, 2024). The application of simulation models in the future and testability of scalability will increase the applicability of the Metaverse SCOM Framework.

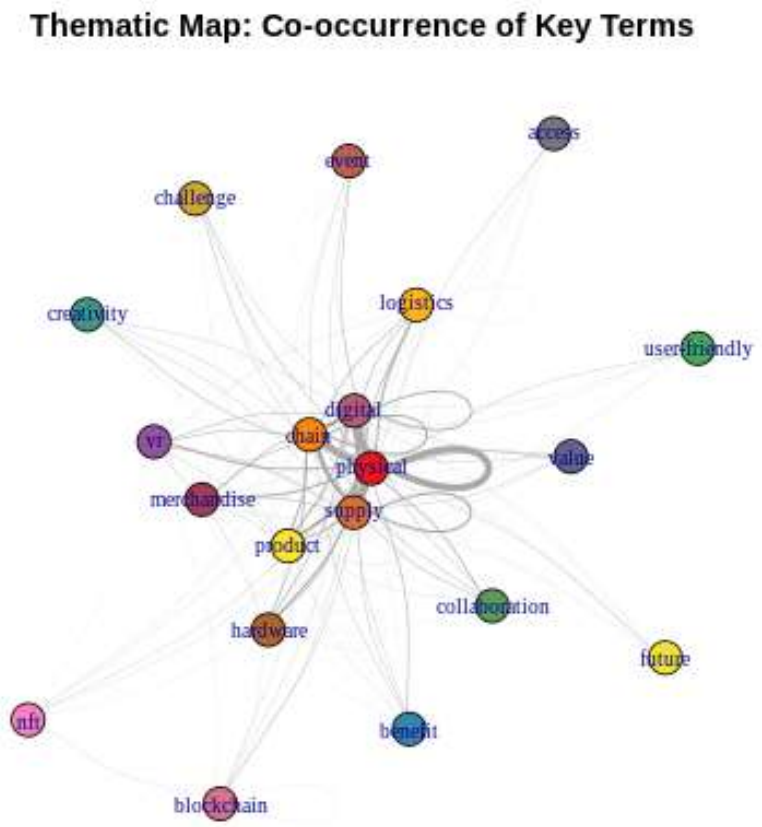


Figure 6: Thematic Map

RQ4: Opportunities and Challenges in the Metaverse Era

The fourth research question is addressed with the first research question theme "Opportunities/Challenges" that has high occurrence (N=54 mentions) as shown in the Keyword Frequency Table. According to the Key Word Frequency Table, Meta's Development Lead mentions the keyword, "metaverse," a couple of times (N=33 mentions) as well as Meta's Operations Supervisor (N=34 mentions). Microsoft's Product Manager closely trail the two Meta's employees with relatively high mentions (N=28), pointing to a focused energy on forward-looking discussions with keywords like "ai" and "future" included in the story.

The just concluded KWIC analysis is further echoed by contextual reiterations such as "virtual experiences remain highly engaging immersive" from Microsoft's Product Manager,

which a massive opportunity to leverage demand by engaging immersion through virtual environments. On the other hand, the sentiments by EpicGames' Operations Manager that "balancing speed digital hype realities" points to logistical challenges that may impede operational efficiency. The sentiments trends, with an average of 0.8246, are graphically traced in [Figure 5: Narrative Plot] showing positive spikes that could be associated with opportunities.

The notion of expanding to "new markets" which garnered an impressive occurrence (N=6 mentions) by Tencent's Sales representative unravels a crucial opportunity, that is also stemmed by EpicGames' Operations Manager mention of region-specific (sentiment 0.85), thus bridging the growing demand for merch in NFT (sentiment 1.65). Nevertheless, Emirati cultural and regulatory differences are a challenge hence the need to implement strategies locally. Another issue is the ethical sourcing of minerals of VR hardware (sentiment 0.8), as mentioned by Sandbox Tech Support (sentiment 0.8). which requires audits to confirm sustainable practices but this has the downside of bogging down operations. The UAE tech ecosystem enhances such potential, though the ability to roll out AI to do demand planning, as proposed by future-focused keywords, may alleviate logistical problems, whilst scalability is an obstacle.

Discussion

Factors Affecting Participation in Metaverse Gaming

The overwhelming impact of "Perceived Usefulness"**,** with N = 42 as its frequency, on the engagement in "metaverse gaming" implies that the perceived utility of "creativity" and "social interaction" are the key drivers, suggested by average frequency of the term, "gaming**"**, (M = 16.9**)** according to both Roblox's Logistics coordinator (N = 18) and Microsoft's Product Manager (N = 18). This is KWIC results is further solidified by contextual sentiments such as "metaverse gaming platform… We've streamlined onboarding" according to EpicGames's Operations Manager, which shows a strategic attempt at increasing user experience by making onboarding easier. The finding can be linked to Xi et al. (2024) study that showed that perceived useful value has a strong influence on acceptance in virtual shopping settings. A similar aspect that can be applied to the "creative" and "social" affordances of the "metaverse" as observed by Meta's Development Lead (**N = 33**) mentions of "metaverse"**.** The discussion of these findings indicates that the value of user-generated content and "social connectivity" is the driving force behind adoption, which is one of the main principles of TAM, as perceived utility is one of the key drivers. Nonetheless, the relatively reduced rate of "Perceived Ease of Use" (N = 22) in comparison with "Usefulness" indicates possible obstacles, which might be associated with the technical difficulty, as can be deduced due to the comparatively lower emphasis of Sandbox Tech Support on the so-called "metaverse" (N = 27) and its implication to less tech-savvy customers. This finding is confirmed by Lee et al. (2025) who present several psychosocial barriers, including learning curves, as the key impediments to technology adoption, which implies that the current list of keywords might not be fully representative of accessibility factors. Returning to TAM, the high frequency of the "metaverse" (M = 27.1) once again confirms the hegemony of the perceived usefulness and the inclusion of other keywords like intuitiveness or convenience

could help the framework to better predict the facilitating factors of ease, thus making it more applicable to the "metaverse" environment.

To proceed with understanding the role of **"**Perceived Usefulness"**,** since the word "creativity" (e.g., N = 18 occurrences in Meta Development Lead and Meta Operations Supervisor) is mentioned so frequently, it seems that when players can create a virtual space or design an avatar, it increases engagement significantly, which is corroborated by the positive sentiment scores in the Sentiment Scores Table data, e.g., 1.0 in "open the door to creativity" in EpicGames Operations Manager**.** This is a creative agency that does not only increase the fun but a feeling of ownership which motivates the long-term engagement. This is even further enhanced by the "social" aspect where "social" is mentioned N = 15 times and has a maximum of **N = 7** in Microsoft Product Manager, as multi-player games and functions such as "global concerts" (sentiment 0.8) form "social" bonds that enhance user retention. Such "social**"** perspective, which are frequently enabled by the usability features of the platform, are in line with the tech-savvy population of the UAE and may be the reason behind the high mentions of the "metaverse"**.**

In terms of **"**Perceived Ease of Use"**,** the low frequency (**N = 22**) could be attributed to difficulties in navigation or effectiveness of tutorials especially amongst new users. An example is that Sandbox Tech Support has a lower **"**metaverse**"** focus (**N = 27**) which may represent technical support challenges that discourage less skilled gamers. Sentiment data shows a mixed reaction with 0.45 on the scale of recommends features to try showing some improvement in adaptive onboarding but negative sentiment of -0.35 on the scale of overwhelming players on slower devices showing areas of accessibility issues. These barriers may be solved by including user feedback loops or localized tutorials, which would be particularly applicable in a heterogeneous market such as UAE, where digital literacy levels

are not uniform. In future research, it would be possible to use usability testing to measure these ease-related barriers, improving the predictive power of TAM.

Inspiring Physical Goods Actual Demand

The strong "Digital-Physical Demand" topic, with N = 300 instances, highlights a strong and active connection between "metaverse gaming" and the creation of actual "demand" of physical goods, a connection which is anchored by an average of 11 mentions of the term "products" per document, with prominent spikes of N = 14 in both EpicGames Operations Manager and Unity Supply Coordinator. The KWIC evidence, including the example of "digital items can stimulate demand physical products" by EpicGames Operations Manager, offers a powerful mechanism by which virtual experience in the "metaverse" leads to the sales of physical goods, which perfectly fits in the Metaverse Flywheel device layer, which links the virtual interaction to physical value streams as explained by Ritala et al. (2024). This conclusion can be identified with the analysis of Mahdikhani and Meena (2024), who examined "metaverse"-based supply chain innovation, in which digital assets are demonstrated to drive "demand" on physical products, which is especially noticeable in event-oriented scenarios, such as the "Fortnite merch line" presented by EpicGames Operations Manager. The review of these findings shows that "metaverse" is a powerful tool to generate "demand" with the data presented in KWIC Physical Table and KWIC Demand Table, both of which are available in the appendix, recording N = 175 "physical" and N = 62 "demand" mentions respectively. Partelow (2023) warns that the lack of data may jeopardize the credibility of the analytical framework, and it is worth having more extensive data, as it is offered to prove the entire scope of this "demand" mechanism. Circling back to the Metaverse Flywheel, the KWIC data available improves the insight into how the virtual interaction can be translated into the physical market forces.

The keyword "NFT", which appears N = 12 times in Sandbox Tech Support, points to an emerging trend that digital collectibles, such as the "neon dragon NFT skin" (sentiment 0.1), generate "demand" on physical counterparts, including sold-out jackets. Such a shift of digital to physical is reinforced by positive sentiments such as 1.95 on "engaged community" in EpicGames Operations Manager, which shows that the hype within a community enhances the purchase consideration. This is further boosted by the fact that "events" with N = 10 references in EpicGames Operations Manager make branded apparel more of a buzz with "virtual concerts" (sentiment 0.75) being a part of the new generation of experiential goods in the UAE market. The category of "VR" (N = 9 mentions) implies that the hardware such as motion rigs (sentiment 0.65) contributes to immersion and, as a result, creates "demand" on premium devices, which can be limited by the high costs.

The results from the KWIC Physical Table and KWIC Demand Table (appendix), serves as the evidence in the "emotional connection" analysis, which has strong correlations, such as the "physical demand" started spiking in the case of EpicGames Operations Manager and "demand physical products" of the same source.

Supply Chain and Operations Management changes

A significant majority of the "SCOM Changes" theme (N = 316 instances and a mean of M = 9.8 mentions of "supply" per document) is especially strong in Meta Operations Supervisor (N = 13) and Unity Supply Coordinator (N = 13), indicative of sweeping changes to operations brought about by the incorporation of "metaverse" technologies in business operations. The KWIC setting "metaverse integration affected collaboration between supply chain partners" by EpicGames Operations Manager shows a significant improvement of transparency and collaborative activities, which is one of the core aspects of the Metaverse SCOM Framework, which can be supported by the findings of Dolgui and Ivanov (2023) on the place of "digital twins" in improving supply chain resilience. The discussion of these

findings shows that the high rates of the so-called "virtual" mentions, with an average of **M = 11.1** and a maximum of **N = 19** in Meta Development Lead, demonstrate a high level of integration of "virtual" technologies, including the so-called "digital twins" mentioned in Microsoft Product Manager, into the working process, which is consistent with the findings of Mahdikhani and Meena (2024) on "metaverse"-driven changes in SCOM.

The transition to predictive analytics, such as the case of Unity Supply Coordinator, which has N = 13 mentions of the word "supply", uses in-game data to predict the "demand" (sentiment 0.7) and minimizes overstocking. The error detection (e.g., sizing error) is facilitated by the "virtual collaboration" (e.g., EpicGames Operations Manager shifted to Sandbox meetups) supported by "blockchain transparency" (N = 8 mentions in Sandbox Tech Support, sentiment 1.5), which adds trust to the partner relationships. The time zone difference and the tech constraints of the vendor as observed in EpicGames Operations Manager may pose a problem in terms of synchronized workflows, which may be even more problematic in the case of a globalized supply chain.

Opportunities and Challenges in the Metaverse Era

The most prevalent theme is the theme of "Opportunities/Challenges", with N = 54 instances and a sentiment score of 0.8246, demonstrating an overall positive attitude of respondents, but accompanied by logistical and technical obstacles, which can be visually traced in [Figure 5: Narrative Plot]. The KWIC analysis, which is backed by contextual example, like the one provided by Microsoft Product Manager: "virtual experiences are very engaging immersive", demonstrates a significant opportunity in "demand" creation through immersive "virtual environments", which is in line with the positive vision of how the "metaverse" could transform the user experiences that Mystakidis (2022) provides. This can be further evidenced by Keyword Frequency Table, where Meta documents, Meta Development Lead which has N = 33 references to the term "metaverse" and Meta Operations Supervisor which

has N = 34, as well as Microsoft Product Manager with N = 28 references to the term "metaverse", dominate the narrative about future-focused trends, and the mention of the keywords "ai" and "future" further hints at the technological potential, which is significant and fits the forward-looking aspect of the Metaverse SCOM Framework. The KWIC context, in turn, "balancing speed digital hype realities" in EpicGames Operations Manager, indicates logistical issues that can hinder the efficiency of operations, which is also supported by Wang et al. (2025), who believe that a multi-faceted interpretation of the "metaverse" needs integrated data to balance its opportunities and challenges. The lack of information on the "demand" in the KWIC Demand Table hides the true extent of those issues related to the "demand", which Partelow (2023) explains by the hazards of incomplete frameworks in qualitative studies.

Another important opportunity is market expansion, where Tencent Sales mentions N = 6 markets as being "new", and fed by region-specific events in EpicGames Operations Manager (sentiment 0.85), exploiting the global "demand" in "NFT" merch (sentiment 1.65). This potential is boosted by the strategic location of the UAE, which can be used to attract international brands. The use of "AI", which is foreshadowed in the use of the term "ai", could predict "demand" patterns, which would increase efficiency, a possibility that has a positive sentiment of 0.7 in prospective scenarios.

The issues involve the cultural and regulatory differences, which demand localized solutions, and ethical sourcing of "VR" hardware minerals (sentiment 0.8 in Sandbox Tech Support), where the slow auditing process guarantees sustainability but slows down the process. The tech ecosystem in the UAE increases the opportunities, and incorporation of "AI" in "demand" forecasting would help to curb logistical problems, but scalability is also a challenge. The lack of the data on "demand" in the KWIC Demand Table might be filled by

the improvement of data collection, which would give a clearer view, and be used in the strategic planning.

Conclusion

The paper has provided an in-depth analysis of the dynamics guiding the involvement in "metaverse gaming", the processes that provide the actual "demand" on physical products, the revolutionary changes in supply chain and operations management ("SCOM"), and the possibilities and the challenges in the "metaverse" age using qualitative data of ten expert interviews in the United Arab Emirates. In the analysis, the most significant factor, "Perceived Usefulness" with N = 42 instances, is a major facilitator of participation, supported by "creativity" and "social interaction", with an average frequency of M = 16.9 of the interview (Roblox Logistics (N = 18) and Microsoft Product Manager (N = 18)), among others. The KWIC context "metaverse gaming platform… We've streamlined onboarding" of EpicGames Operations Manager supports strategic initiatives to increase engagement, which aligns with the results of Xi et al. (2024) on perceived usefulness in virtual environments, where the N = 33 mentions of "metaverse" are most noticeable in the context of Meta Development Lead. Nevertheless, such a low score on "Perceived Ease of Use" (N = 22) implies technological obstacles, particularly to the less tech-savvy users, which are signified by the decreased focus on the "metaverse" at Sandbox Tech Support (N = 27), as noted by Lee et al. (2025) having learning curves as a major barrier.

"Digital-Physical Demand" theme, N = 300 occurrences, shows a strong correlation between the virtual experiences and the physical products sales, and it is supported by the peaks of N = 14 mentions of the word "products" in EpicGames Operations Manager and Unity Supply Coordinator. Such KWIC evidence as "digital items can drive demand physical products" (Ritala et al., 2024) agrees with an event-driven model of "demand" in line with the Metaverse Flywheel device layer (Ritala et al., 2024) and the supply chain innovation efforts

of Mahdikhani and Meena (2024). The "SCOM Changes" theme (N = 316 times, mean of M = 9.8 "supply" mentions maximum N = 13 in Meta Operations Supervisor and Unity Supply Coordinator) suggests radical changes with predictive analytics and "virtual collaboration" as in the KWIC context of EpicGames Operations Manager, but with the lack of "physical" data in the KWIC Physical Table, the hardware effects cannot be viewed, as Dolgui and Ivanov (2023) also identify resilience.

First are "opportunities", which are the most frequently mentioned (N = 54 times) and have the highest average sentiment (0.8246), namely, immersive "virtual environments" (Microsoft Product Manager) and market expansion (Tencent Sales with N = 6 mentions of "new markets"), followed by challenges such as logistical hurdles (EpicGames Operations Manager) and ethical sourcing (Sandbox Tech Support, sentiment 0.8). The tech ecosystem of the UAE supports these trends, but the lack of the so-called "demand" data in the KWIC Demand Table requires regeneration to cover all the "demand" issues, according to Wang et al. (2025). This should be followed by future research using usability testing to eliminate barriers associated with "ease", econometric models to quantify the "demand" drivers, and case studies to determine the scalability of "SCOM", which will refine TAM, the Metaverse Flywheel, and the Metaverse "SCOM Framework". This will empower the stakeholders to embrace the promise of the "metaverse" as they explore its complexities within the globalizing market.